

When Accessibility Becomes a Trap: A User-Centric Characterization of Dark Patterns Arising from Screen Reader Users' Perceived Deception in Mobile Interfaces

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Abstract

Dark patterns manipulate users into making choices they might not otherwise make by leveraging interface-level control. While research on dark patterns has grown in recent years, most studies focus on visually deceptive designs and how they intentionally mislead users. However, little attention has been paid to the misleading experiences of BLV users who rely on screen readers, particularly those arising from accessibility barriers and their unique perceptual mechanisms in mobile interactions. To address this gap, our study adopts a user-centered approach that focuses on the lived experiences of screen reader users, identifying ten mobile-specific types of dark patterns, including patterns unique to the BLV experience. Furthermore, we propose the SCAR framework to provide systematic guidance for identifying and mitigating dark patterns, aiming to advance the fairness of mobile accessibility and promote greater industry accountability.

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CSCW Companion '25, Bergen, Norway
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ACM ISBN 979-8-4007-1480-1/25/10
<https://doi.org/10.1145/3715070.3749249>

CCS Concepts

• **Human-centered computing** → **Accessibility theory, concepts and paradigms.**

Keywords

Dark Patterns, Deceptive Design, Blind and Low-Vision Users, Mobile Accessibility, User-Centric Taxonomy, Accessibility Barriers, Screen Readers

ACM Reference Format:

Dai Shi, Lin Tian, Jiaxun Sun, Tomomi Kawakami, Nuo Cheng, Shuchang Xu, and Guanhong Liu. 2025. When Accessibility Becomes a Trap: A User-Centric Characterization of Dark Patterns Arising from Screen Reader Users' Perceived Deception in Mobile Interfaces. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3715070.3749249>

1 Introduction

Dark patterns, interface design strategies manipulating user decisions via systematic misdirection or coercion [1], are common in scenarios like e-commerce checkouts and privacy authorizations, often causing user harm. While extensively discussed [2, 3], existing research largely adopts visual interaction paradigms, neglecting the specific experiences of blind and low vision (BLV) users. Due to their primary reliance on screen readers, BLV users' unique perceptual and operational pathways may amplify the privacy, financial, and cognitive risks posed by dark patterns [3–5].

Lewis et al. [6] takes an initial step to investigate how BLV users encounter dark patterns, revealing that—in addition to intentionally deceptive designs—accessibility barriers can also create a sense of manipulation. However, current dark pattern frameworks are limited in scope: they primarily focus on visually deceptive interfaces designed to mislead sighted users, neglecting *how BLV users interpret designs as deceptive*. Consequently, these frameworks inadequately represent the digital manipulation risks encountered by BLV users, highlighting the need to refine and expand them through the lens of BLV users' lived experiences.

To bridge this gap, we conducted an interview study with 11 BLV screen reader participants to investigate how they perceive and encounter deceptive designs and dark patterns in mobile interfaces. Participants shared detailed narratives and demonstrations of past experiences where they felt misled or deceived by interface designs. Through systematic analysis guided by the four WCAG principles—Perceivable, Operable, Understandable, and Robust—we identified [7] **ten deceptive patterns affecting BLV users** (e.g., Focus Hijacking, Label Deprivation). Our findings demonstrate that accessibility barriers not only limit functionality but *actively create deceptive experiences by disrupting users' cognitive processes*—such as when missing or misleading element labels induce navigation disorientation and erode trust in the interface.

To address these issues, we further propose the **SCAR framework**, a set of guardrails designed to prevent accessibility-related deception. This framework operationalizes key interaction metrics, including focus purity and rhythm synchronization, to quantify deceptive risks. Our work underscores the urgent need for both researchers and practitioners to expand conventional definitions of dark patterns to include accessibility-related deception, ensuring more inclusive digital experiences for users with accessibility needs.

2 Related Work

2.1 Deceptive Design: Taxonomies and Scope

The concept of 'Dark Patterns,' introduced by Brignull [1] to describe interface strategies that manipulate user decisions via misdirection or coercion, pertains to prevalent designs widely recognized for causing harms such as financial loss and privacy breaches [8, 9]. Subsequent academic frameworks, such as those by Gray et al. [10] and Mathur et al. [2], have categorized these manipulations, primarily based on visual presentation features and inferred 'designer intent.' However, the scope of these mainstream taxonomies reveals limitations when applied to BLV users, who have fundamentally different perceptual modalities. Initial research by Lewis et al. [6] indicates that BLV users often experience 'accessibility barriers' as subjective manipulation or deception. Such 'deception-like' experiences, stemming from unintentional design flaws and perceived through non-visual interaction (e.g., auditory cues, interaction flows), differ significantly in origin and manifestation from the visually-focused, intent-driven patterns typically addressed by existing classifications. Consequently, these established frameworks struggle to effectively encompass these unique experiences, highlighting an insufficiency in their inclusive scope.

2.2 Accessibility Barriers and Their Perceptual Consequences for BLV Users

In mobile interactions, Blind and Low Vision (BLV) users relying on screen readers acquire information linearly and primarily through auditory channels [11–13], making them strongly reliant on coherent interface structures, semantic accuracy, and stable interaction rhythms [14, 15]. However, prevalent accessibility barriers in mobile interfaces—such as missing labels, erratic focus management, and inadequate dynamic feedback—extend beyond mere usability issues. These deficits directly disrupt BLV users' perceptual and cognitive processes: labeling problems force guesswork and trial-and-error, leading to misunderstanding and uncertainty [16]; disordered focus and interaction flows can cause disorientation, loss of control, and frustration [17]; and missing dynamic content feedback often results in confusion and inefficiency. Such profound perceptual consequences severely impair BLV users' ability to understand and effectively interact with interfaces, warranting further examination of their nature [10].

2.3 Limitations of Accessibility Guidelines in Mitigating Adverse Perceptual Experiences for BLV Users

While accessibility barriers significantly challenge BLV user experiences, the efficacy of existing accessibility guidelines (e.g., WCAG) in systematically mitigating these deep adverse perceptual experiences is critical. WCAG, a cornerstone for basic digital accessibility, provides vital technical specifications via its POUR principles to ensure assistive technologies can reach information and functionalities [7]. However, these guidelines primarily focus on element-level technical compliance, such as ensuring alt text for images or keyboard operability for controls. For complex dynamic interactions or scenarios requiring precise multi-sensory information synchronization, WCAG's guidance is relatively high-level, potentially failing to prevent cognitive confusion arising from disrupted interaction rhythms or semantic ambiguities [14]. Crucially, WCAG's core aim is "accessibility," not directly preventing or identifying patterns where (potentially unintentional) accessibility deficits functionally equate to "deception" or "manipulation" in BLV users' subjective experience. Thus, a guidance gap persists in systematically helping designers avoid such profound negative experiences.

3 Methods

We conducted semi-structured interviews with 11 BLV participants (P1–P11). Among them, 7 participants were totally blind, and 4 had residual light perception. We recruited them through social media platforms and word of mouth. Our specific inclusion criteria included: (1) at least 18 years old; (2) a regular user of mobile devices for accessing online services; (3) having basic or advanced proficiency with screen readers. Among the 11 Participants, 6 primarily relied on VoiceOver on iOS devices, while 5 reported utilizing Talkback or third-party screen reader applications on Android mobile devices. Detailed participant information is summarized in Appendix 1.

The interviews began with participants' self-reports on their disability and screen reader usage experience. We then introduced

the overall concept of dark patterns through probing questions and specific experiences (e.g., “Have you ever felt that a design was deliberately pushing or cheating you to do something you didn’t want to do?”). Participants were then asked to share the types of harm they had encountered and any strategies to help cope with dark patterns. Each interview was conducted online and lasted approximately 30 minutes, with all participants receiving compensation.

4 Results

Unlike existing taxonomies, our approach is user-centered, emphasizing the authentic experiences and perspectives of users when encountering such patterns. It is informed by the four principles of the Web Content Accessibility Guidelines (WCAG) [7]—Perceivable, Operable, Understandable, and Robust—which address access foundations particular for people with disabilities. Through thematic analysis [18] of the interviews, the theme “scenarios and perspectives with dark patterns” was identified and mapped to the four WCAG principles.

Finally we identified ten types of mobile dark patterns affecting BLV users (as shown in Figure 1). Notably, some types arise from the design’s inherent deceptiveness (e.g., Hierarchical Concealment, Mis-trigger Design), while others arise from accessibility barriers that create a sense of deception (e.g., Label Deprivation, Compatibility Deficiency). These latter cases are insightful and often overlooked in existing taxonomies. Detailed results are presented in Appendix 2.

(1) **Focus Hijacking.** Focus hijacking refers to interface elements that divert user focus and disrupt the expected navigation path. It requires them to swipe through advertisements (P2,P3) or pop-ups (P6,P8) before accessing the core functionality. P6 described, “Ads suddenly burst onto the screen mid-playback, aggressively interrupting the content and demanding my full attention.”

(2) **Label Deprivation.** Label deprivation refers to interface elements that lack effective accessibility labels for screen readers. All participants (11/11) highlighted that the absence of such labels amplifies the detrimental effects of Dark Patterns. They were less able to discern misleading content and “*accidentally fall into traps they didn’t anticipate*” (P3).

(3) **Auditory Interference.** Auditory interference refers to elements with auto-playing sounds that interfere with screen reader audio output. This issue is particularly common in pop-up ads (P4,P8), which often trigger sudden and overlap with screen reader playback. As P8 described, “*During that noisy period, I struggle helplessly to find a way to turn them off.*”

(4) **Hierarchical Concealment.** Hierarchical concealment refers to the pattern of hiding critical features deep within multi-layered structures, which impairs navigation efficiency. This forces BLV users, who depend on linear navigation, to “*explore the interface repetitively in order*” (P6) to complete tasks that would otherwise be straightforward.

(5) **Function Blocking.** Function blocking refers to unauthorized features that are visually grayed out without providing accessible explanation. This severely restricts functional discoverability and operational clarity for BLV users. P11 recalled, “*I could tell some*

features existed, but the screen reader didn’t explain why I couldn’t use them.”

(6) **Linguistic Manipulation.** Linguistic manipulation refers to semantically ambiguous or misleading descriptions that cause misinterpretation. Its most common forms include deceptive advertising (P8), hidden terms and conditions (P2, P8), and inducement to make purchases (P5, P10). Additionally, some forms of linguistic manipulation are tied to default selection mechanisms.

(7) **Mis-trigger Design.** Mis-trigger design refers to abrupt or delayed elements may lead to confusion and unintended actions. More than half of the participants (6/11) reported experiencing such dark patterns, with splash-screen advertisements and pop-ups ads being the most frequently mentioned barriers, “*I just slightly shook my phone, without any tapping, and the interface switched*” (P3).

(8) **Exit Maze.** Exit maze refers to convoluted procedures—such as adding multi-layers of interaction, requiring multiple confirmations, or designing complex workflows—that hinder users’ target actions. BLV users, who tend to prefer simple and straightforward interfaces, often face difficulties in completing tasks such as unsubscribing (P2, P8), canceling payments (P3), or deregistering their accounts (P5).

(9) **Path and Time Constraints.** Path and time constraints refer to constrained paths or strict time limits that mislead to unintended actions. Path constraints mainly include automatic redirection after a countdown (P5) or compulsory switching to a screen reader mode upon detecting screen reading softwares (P7), which diminish their autonomy in navigating the interface. Time constraints, such as creating the illusion of a limited-time offer (P2), may compel them to make decisions without adequate deliberation.

(10) **Compatibility Deficiency.** Compatibility deficiency refers to dynamic interface changes or updates that are incompatible with screen readers. Common issues include accessibility regression, “*After a software update, more barriers appeared*” (P2). Another issue arises from interference caused by asynchronous content. As P3 mentioned, “*Frequently popping ads lack an accessible close button, making it difficult for screen readers to identify such dynamic content.*”

5 Design Consideration

The 10 types of BLV-specific dark patterns summarized in Figure 1 demonstrate how mobile interfaces can disrupt screen-reader-centric cognitive workflows for BLV users by hijacking the initial focus or audio track, embedding hierarchical tunnels within screen-reader sequences, implementing controls prone to mis-trigger, and coercing users with countdowns and repeated confirmations. These disruptive effects are empirically substantiated: focus relocation via screen readers imposes cognitive costs 3–5 times higher than visual scanning; concurrent audio streams significantly elevate cognitive load and error rates; and each additional hierarchical level exponentially increases task completion times. Our interview data corroborate these insights, indicating these manipulative strategies disproportionately amplify along the linear cognitive pathway of auditory perception-semantic interpretation-motor execution that BLV users rely upon.

Current mainstream accessibility standards (e.g., WCAG 2.2 [19]; iOS/Android A11y checklists [20, 21]) define compliance primarily

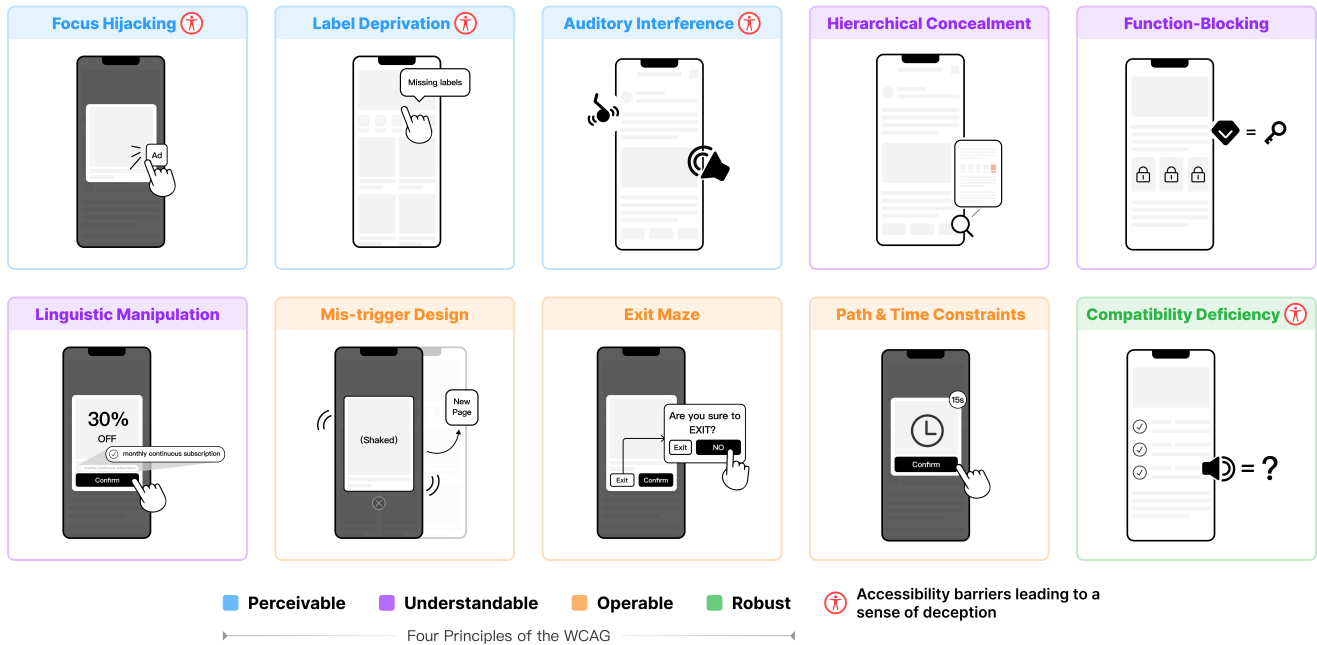


Figure 1: Taxonomies of dark patterns in mobile interfaces of BLV users.

through the reachability of interface elements and the completeness of their labels, implicitly assuming interactional neutrality. However, our observations reveal that designers, even when fully complying with accessibility specifications, can still introduce subtle cognitive biases and indirect misdirection. This is often achieved by manipulating the accessibility layer (e.g., focus order, ARIA semantics) and by exploiting cross-modal translation mechanisms, such as visual-to-auditory or visual-to-haptic mappings. For BLV users, who cannot recalibrate via peripheral vision or holistic interface overview, such deceptive perceptual tactics constitute elusive dark patterns. Such compliant manipulation poses a disproportionate risk which undermines BLVs' autonomy.

The central issue lies in the fact that current accessibility standards have yet to recognize user perception as a distinct evaluative dimension. In the absence of diagnostic frameworks capable of quantifying focus integrity, screen-reader rhythm synchrony, and interaction reversibility, dark patterns are able to pass compliance audits undetected—legally embedded yet structurally corrosive, forming invisible fissures within the accessibility ecosystem.

Precisely because these dark patterns are not solely the result of intentional design, but emerge at the boundary of user perception, we center the BLV experience and propose the SCAR (Sightline–Cognitive–Autonomy–Rhythm) Shield Framework. Framed through the metaphor of shield, SCAR advocates for the establishment of explicit boundaries throughout the mobile application development lifecycle to prevent dark patterns from infiltrating BLV interaction experiences under the guise of ambiguous compliance. The framework outlines four targeted safeguards:

- **Sightline Integrity Shield:** Ensures visual and auditory entry points remain unmanipulated, validating the uniqueness and order of initial focus to maintain stable information intake.
- **Cognitive Transparency Shield:** Requires that control elements disclose intent, cost, and reversibility prior to activation; commercially biased ambiguity is flagged as non-compliant.
- **Autonomy Continuity Shield:** Ensures that termination paths are both discoverable and reversible, with exit flows accessible within no more than three steps to prevent coercive continuity.
- **Rhythm-Sync Resilience Shield:** Regulates unsolicited interface updates, requiring all runtime insertions to emit accessible events to maintain screen-reader and UI synchrony.

These shields can be seamlessly embedded into canonical stages of the product lifecycle—requirements analysis, low-fidelity prototyping, implementation, quality assurance, and post-deployment monitoring (see Figure 2). Rather than overhauling existing workflows, SCAR offers a set of quantifiable and verifiable constraints that expose latent risks and enhance accountability at each phase.

We argue that employing the SCAR framework as a protective boundary for BLV users facilitates the systematic identification of interface patterns that may induce a sense of manipulation. This approach helps to intercept structural exploitation of digitally vulnerable groups at its source. In contrast to the prevailing baseline of mere element reachability, SCAR introduces a longitudinal intervention mechanism spanning the product lifecycle—helping teams embed respect for user rights directly into both design logic and delivery pipelines without compromising development efficiency.

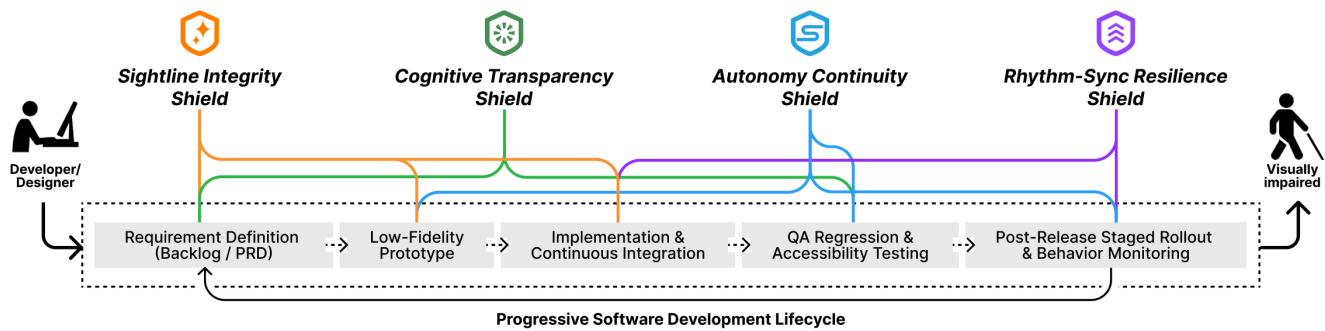


Figure 2: The SCAR Shield Framework.

6 Limitations and Future Work

A user-centered reclassification of ten dark pattern types for BLV users was developed to capture manipulation experiences specific to this population. However, the findings indicate that such experiences may also result from common accessibility barriers, which disrupt BLV users' understanding of interface structures and interaction intent. These barriers can make non-deceptive designs appear manipulative, a phenomenon underexplored in existing research. Future work should investigate how accessibility barriers influence manipulation perceptions across different cognitive and sensory contexts.

The SCAR framework proposed in this work is based on participants from a specific cultural and technological context, raising questions about its generalizability to other regions, languages, and assistive technologies. Its practical effectiveness, usability, and system compatibility also remain untested. Future research should validate the framework with more diverse users and real-world applications, supported by collaboration among designers, developers, and user communities to advance accessible and trustworthy digital experiences for BLV users.

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A Appendix

Table 1: Basic information of interview participants

ID	Age	Gender	Mobile Types	Screen Readers Usage	Disability	Onset
P1	35	Male	iOS	Basic Proficiency	Blind	Acquired
P2	24	Female	iOS	Advanced Proficiency	Blind	Acquired
P3	29	Male	Android	Basic Proficiency	Blind	Acquired
P4	36	Male	Android	Advanced Proficiency	Light Perception	Acquired
P5	36	Male	Android	Advanced Proficiency	Blind	Congenital
P6	27	Female	Android	Basic Proficiency	Light Perception	Congenital
P7	37	Male	Android	Advanced Proficiency	Blind	Acquired
P8	30	Female	iOS	Advanced Proficiency	Light Perception	Congenital
P9	25	Female	iOS	Basic Proficiency	Blind	Congenital
P10	41	Female	iOS	Basic Proficiency	Light Perception	Acquired
P11	28	Male	iOS	Advanced Proficiency	Blind	Acquired

Table 2: Dark patterns in mobile interfaces: types and scenarios for blind users

Principles	Types	Descriptions
Perceivable	Focus Hijacking	Elements that hijacking interface focus disrupt expected navigation path.
	Label Deprivation	Interface elements lack effective labels for screen readers.
	Auditory Interference	Elements with auto - playing sounds interfere with screen reader audio output.
Understandable	Hierarchical Concealment	Hiding critical features in multi - layered structures impairs navigation efficiency.
	Function - Blocking	Unauthorized features are visually grayed out without providing accessible explanation.
	Linguistic Manipulation	Semantically ambiguous or misleading descriptions cause misinterpretation.
Operable	Mis - trigger Design	Abrupt or delayed elements may lead to confusion and unintended actions.
	Exit Maze	Convolutd procedures for unsubscribing or exiting hinder target actions.
	Path & Time Constraints	Constrained paths or strict time limits mislead to unintended actions.
Robust	Compatibility Deficiency	Dynamic interface changes or updates are incompatible with screen readers.